Kahaan Padiya

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Education

Pandit Deendayal Energy University — Gandhinagar, Gujarat
B.Tech in Electronics & Communication Engineering — *Class of 2027*

- **CGPA:** 8.64 / 10.0
- **Relevant Coursework:** Embedded Systems, Mixed Signal VLSI, Digital Signal Processing, Microprocessors & Microcontrollers, Circuits & Systems, Analog Electronics, Analog & Digital Communications, Machine Learning.

Technical Skills

Hardware & Embedded: AVR (ATmega2560), ARM Cortex-A72 (Raspberry Pi 4), Asus Tinker Board, 8086, ESP32, Sensors
Protocols & Interfacing: UART, SPI, I2C, PWM, GPIO Interrupts, Motor Actuation, ADC/DAC Interfacing
Simulation & EDA: KiCad (PCB Layout), LTspice, Tina-TI, Logisim, Proteus, MATLAB, Simulink, Fusion360
Programming: C/C++ (AVR, Bare-metal/Embedded), Python (NumPy, SciPy), Assembly (8086), Bash
Core Domains: Embedded Systems Firmware, Motion Control & Actuation, Mixed-Signal Design, DSP, Bioacoustics
Soft Skills: Technical Documentation, Research Aptitude, Crisis Management, Project Documentation, Team Coordination

Experience

Research Intern (Embedded Systems & Hardware) — May 2026 – July 2026
IISc Bangalore – Dept. of Instrumentation & Applied Physics (Prof. Sai Siva Gorthi) — *Bengaluru, KA*

- Developed embedded firmware in C++ for an electronic pipette, implementing precision stepper motor motion control, persistent calibration management, serial communication, and automated homing routines.
- Engineered custom **PCBs** in KiCad and modeled **3D-printed enclosures** in CAD, iterating on mechanical clearances, structural ergonomics, and component placement.
- Executed end-to-end electro-mechanical integration and calibration to improve fluid dispensing accuracy, repeatability, and device reliability.

Summer Intern — June 2024
Jivdaya Charitable Trust (Animal Welfare) — *Ahmedabad, Gujarat*

- Assisted in veterinary support and operational logistics for wildlife rehabilitation.
- Contributed to a **raptor research project** by processing clinical and behavioral data for analysis.
- Demonstrated effective crisis management and adaptability in a high-pressure field environment.

Projects

AnyCall v2: Autonomous Bioacoustic Edge Sensor System | *Python, TFLite, Raspberry Pi, DSP* — 2026

- Engineered an end-to-end multi-taxa bioacoustic sensor node pairing an anisotropy-centered 1024-D deep embedding with a deterministic 40-D acoustic morphology descriptor (spectral shape, 2–200 Hz pulse rates, and cepstral stats).
- Developed a calibrated two-view open-set gate achieving **94.38% macro accuracy** across 52 Indian wildlife species and **97.60% selective accuracy** under a strict 5-seed recording-disjoint holdout protocol.
- Profiled on a fanless Raspberry Pi 4 CPU (264 ms latency, **11.4× faster than real-time**) within 555 MB RAM, and ported to a standalone on-device Android TFLite application (Under review at IEEE APSCON 2026).

Minimoog-Inspired Polyphonic Digital Synthesizer | *C++, DSP, Embedded Systems* — 2026

- Developing a **polyphonic Virtual Analog (VA)** synthesis engine in C++, implementing an 8-voice allocation architecture and emulating non-linear transistor ladder filters for real-time embedded performance.

Design & Simulation of Delta-Sigma Modulator | *MATLAB, Simulink* — 2026

- Modeled a **2nd-order Delta-Sigma modulator** in Simulink to analyze noise-shaping for high-fidelity audio.
- Validated the system architecture and signal flow, analyzing trade-offs between Over-Sampling Ratio (OSR) and Signal-to-Noise Ratio (SNR).

MATLAB Innovation Challenge: Code the Future (Winner) | *MATLAB, Simulink* 2025

- Led a team to develop an algorithm that determines the precise solar panel positioning for maximum possible output at any given location.
- Integrated complex multi-variable analysis including annual sun path, shadow casting, atmospheric conditions, and panel thermal properties.

Beam Steering & Antenna Array Optimization Tool | *MATLAB, DSP* 2024

- Designed a simulation tool to visualize and optimize beamforming for antenna arrays.
- Implemented algorithms for phase shifting and array factor calculation; highly relevant for audio spatial applications.

Campus Navigation Tool | *QGIS, Web App* 2023

- Mapped all university buildings using **QGIS** and deployed a web app accessible via QR codes stationed across campus.
- Implemented an autocomplete search feature, allowing users to instantly locate specific academic and administrative buildings.

Teaching Assistant Robot | *Asus Tinker Board, Arduino* 2019

- Collaborated to build a functional robot assistant; integrated **Asus Tinker Board** for Front end and **Arduino** for motor control.
- Designed the electronics subsystem, establishing early hands-on expertise in hardware integration.

Awards, Honors & Patents

Design Patent - High-Gain Fractal Antenna for Enhanced Wireless Communication 2025

First Place Winner - MATLAB Innovation Challenge: Code the Future (IEEE Hackathon) 2025

Semi-Finalist - Tata Club Energy National Quiz 2020

Leadership & Community

Extracurricular Leadership

- **Technical Committee, Brahmand (Space Club):** Collaborated within a team to build a dual-axis sun tracker; specifically responsible for integrating sensors and servo motor control logic.
- **Committee Member, Sorriso (Photography Club):** Maintained visual archives and managed event documentation for **over 2 years**, ensuring comprehensive coverage of university activities.
- **Member, VGA (Cinema Club):** Managed event documentation and visual coverage for cultural events.

Wildlife Conservation & Citizen Science

- **GBBC Campus Organizer:** Coordinator for the Great Backyard Bird Count, managing volunteers and data compilation.
- **Field Researcher:** Participated in bird counts across Ahmedabad, Dholera, and Nalsarovar.
- **Conservation Photography:** Documenting local biodiversity through wildlife photography to support ecological awareness.
- **NSS Volunteer (2023-2024):** Active participant in the National Service Scheme community service initiatives.